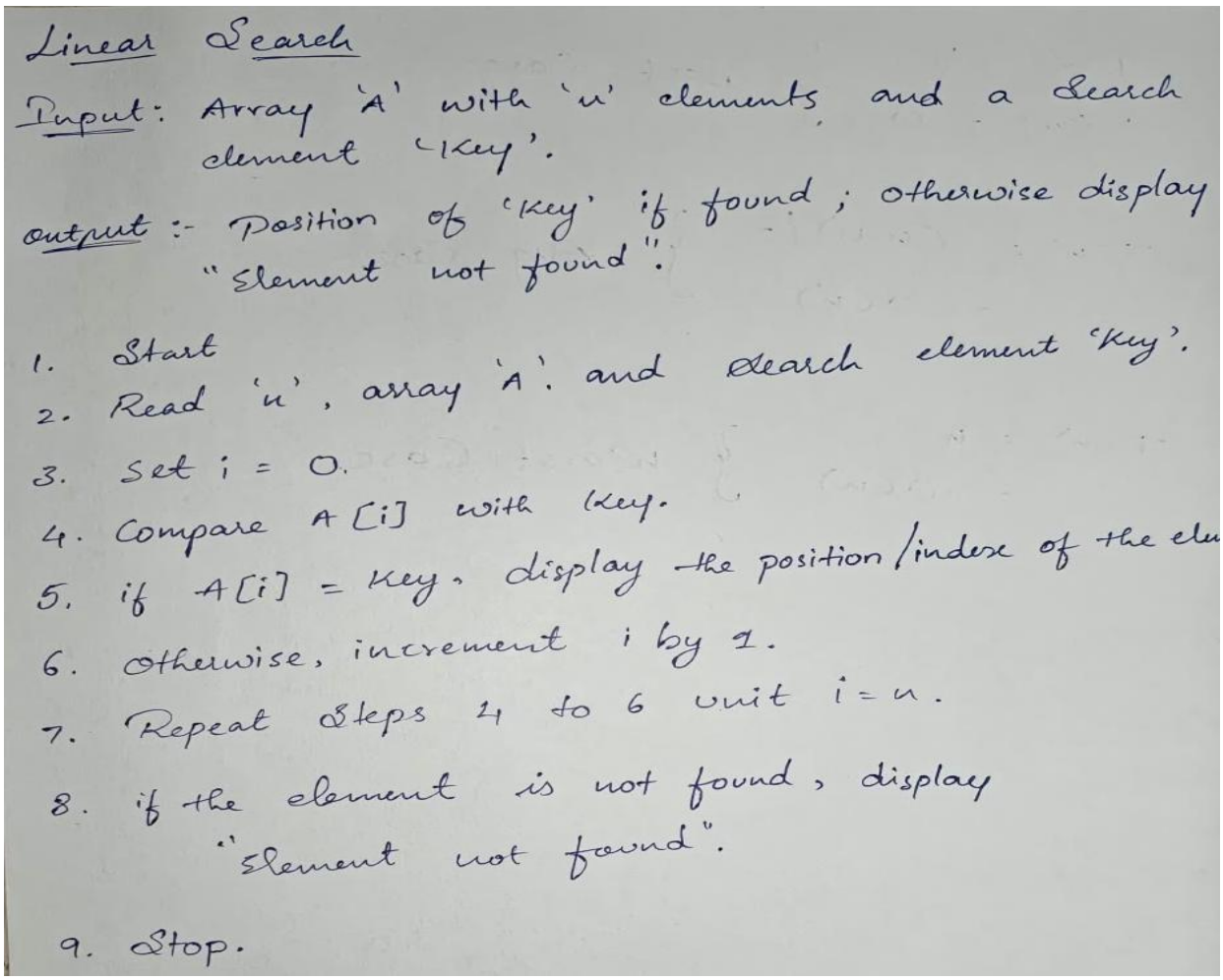
PRACTICAL : 02

Aim:

Implementation and Time analysis of linear and binary search algorithm.

1. Linear Search.

Algorithm:



Linear Search

Input: Array 'A' with 'n' elements and a Search element 'key'.

output :- Position of 'key' if found ; otherwise display "Element not found".

1. Start
2. Read 'n', array 'A' and Search element 'key'.
3. Set i = 0.
4. Compare A[i] with key.
5. if A[i] = key, display the position/index of the ele
6. otherwise, increment i by 1.
7. Repeat steps 4 to 6 unit i = n.
8. if the element is not found, display "Element not found".
9. Stop.

Code:

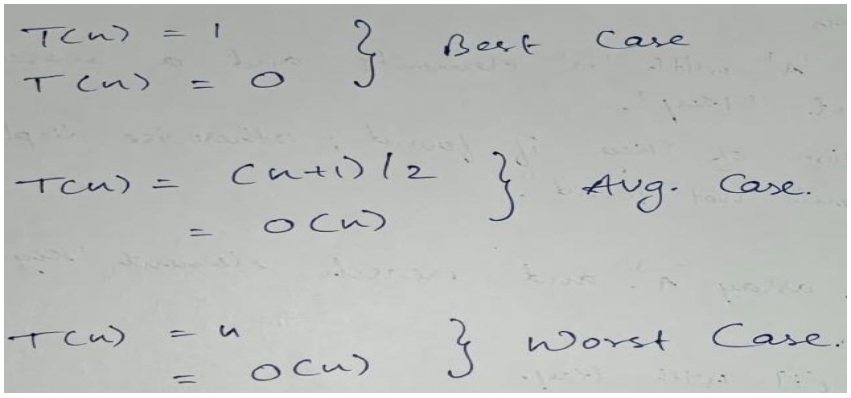
```
#include <iostream>
using namespace std;

int Linearsearch(int arr[], int n, int search)
{
   for (int i = 0; i < n; i++)
   {
      if (arr[i] == search)
      {
         return i;
      }
   }

   return -1;
}

int main()
{
   int n;

   cout << "Enter number of elements: ";
   cin >> n;

   int *arr = new int[n];

   cout << "Enter elements:\n";
   for (int i = 0; i < n; i++)
   {
      cin >> arr[i];
   }

   int search;

   cout << "Enter element to search: ";
```

```
    cin >> search;

    int index = Linearsearch(arr, n, search);

    if (index != -1)
    {
        cout << "Element found at index " << index << endl;
    }
    else
    {
        cout << "Element not found" << endl;
    }

    delete[] arr;

    return 0;
}
```

Output:

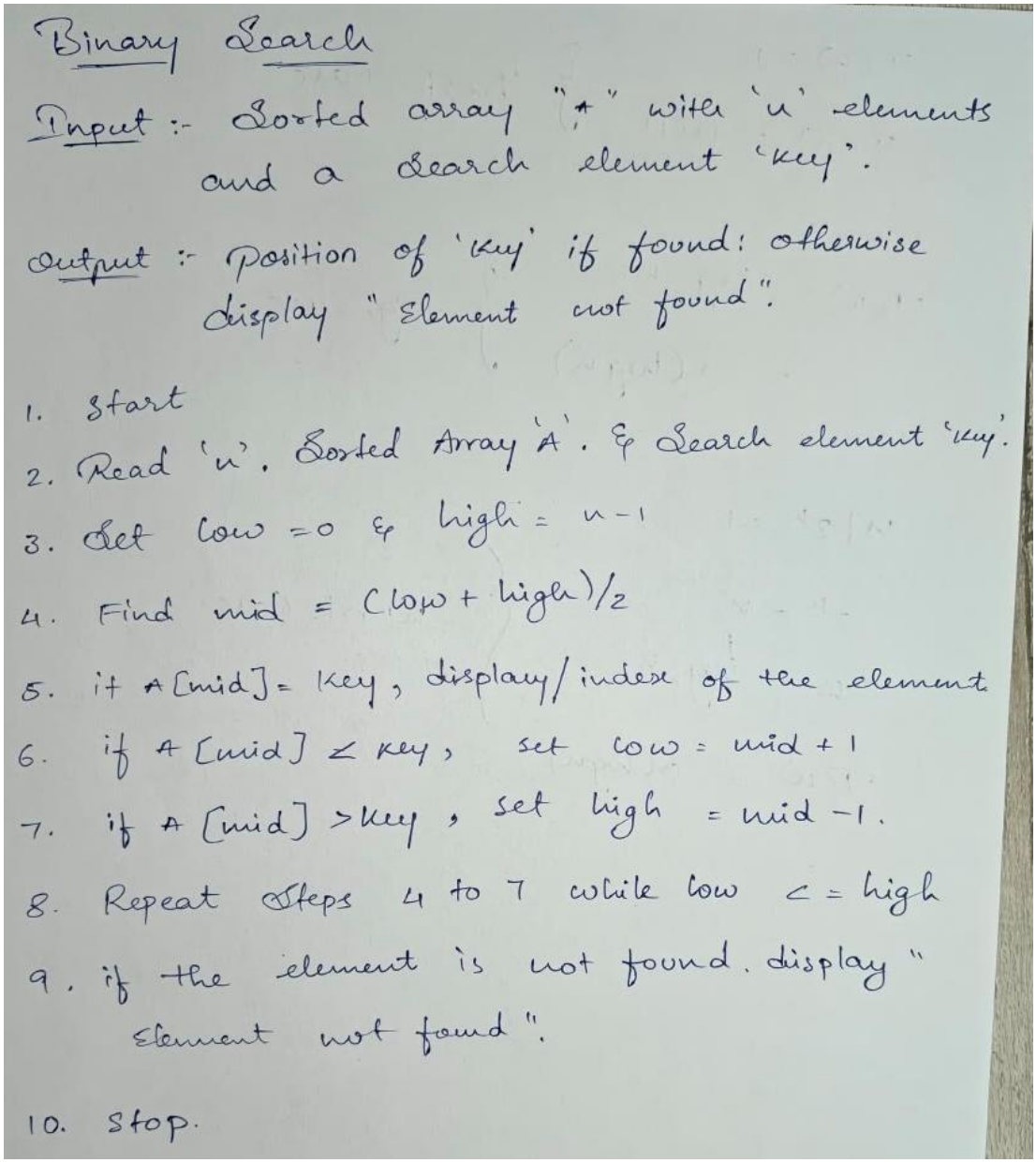
```
Enter number of elements: 6
Enter elements:
2 88 9 4 78 90
Enter element to search: 4
Element found at index 3
```

Time Complexity:



2. Binary Search.

Algorithm:



Binary Search

Input :- Sorted array "A" with 'n' elements and a search element 'key'.

Output :- Position of 'key' if found: otherwise display "Element not found".

1. Start
2. Read 'n', Sorted Array 'A', & Search element 'key'.
3. Set low = 0 & high = n-1
4. Find mid = (low + high)/2
5. if A[mid] = key, display/ index of the element
6. if A[mid] < key, set low = mid + 1
7. if A[mid] > key, set high = mid - 1.
8. Repeat steps 4 to 7 while low <= high
9. if the element is not found, display "Element not found".
10. Stop.

Code:

```
#include <iostream>
using namespace std;

int Binarysearch(int arr[], int n, int search)
{
   int low = 0;
   int high = n - 1;

   while (low <= high)
   {
      int mid = low + (high - low) / 2;

      if (arr[mid] == search)
      {
         return mid;
      }
      else if (arr[mid] < search)
      {
         low = mid + 1;
      }
      else
      {
         high = mid - 1;
      }
   }

   return -1;
}

int main()
{
   int n;

   cout << "Enter number of elements: ";
   cin >> n;
```

```
    int arr[n];

    cout << "Enter elements in sorted order:\n";
    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    int search;

    cout << "Enter element to search: ";
    cin >> search;

    int index = Binarysearch(arr, n, search);

    if (index != -1)
    {
        cout << "Element found at index " << index << endl;
    }
    else
    {
        cout << "Element not found" << endl;
    }

    return 0;
}
```

Output:

```
Enter number of elements: 4
Enter elements in sorted order:
23 44 67 88
Enter element to search: 88
Element found at index 3
```

Time Complexity: